STAT201 · STATISTICS · LECTURE 4

# Random Variables and Discrete Distributions

How to describe an uncertain numerical outcome by a distribution, an expectation and a variance

Syllabus topic 4 · week 4 · 4 academic hours

**LECTURE GOALS**

- distinguish a discrete random variable from a continuous one and specify the probability mass function (PMF) and the cumulative distribution function (CDF);
- compute the expected value, variance and standard deviation of a discrete variable;
- apply the properties of $\mathrm{E}[aX+b]$ and $\mathrm{Var}(aX+b)$ and the addition rule for independent variables;
- recognize situations described by the binomial, geometric and Poisson distributions and compute their probabilities;
- use the expected value and risk (standard deviation) to compare decisions.

## 1 A random variable and its distribution

In Lectures 2–3 we worked with events. Most often an outcome is expressed by a number: the profit on a loan, the number of calls per hour, the number of defective parts in a batch. This is conveniently described by a function that assigns a number to each outcome.

**RANDOM VARIABLE**

A random variable $X$ is a numerical function of the outcome of a random experiment. Values are denoted by lowercase letters: $\{X = x\}$ is the event "the variable $X$ took the value $x$".

A random variable is *discrete* if it takes a finite or countable number of values (the number of customers, the number of defects). It is *continuous* if it takes any values on an interval (height, waiting time, an exchange rate). This lecture is devoted to discrete variables; continuous ones are covered in Lecture 5.

**PROBABILITY MASS FUNCTION (PMF)**

For a discrete variable the function $p(x) = P(X = x)$ is called the probability mass function. It has the properties

$$p(x) \geq 0, \qquad \sum_{x} p(x) = 1. \tag{1.1}$$

**CUMULATIVE DISTRIBUTION FUNCTION (CDF)**

$F(x) = P(X \le x) = \sum_{t \le x} p(t)$. This is a non-decreasing "step" function from 0 to 1. For integer values $P(a < X \le b) = F(b) - F(a)$.

**EXAMPLE 1. CALLS TO A SUPPORT LINE**

The number of contacts $X$ in 10 minutes has the distribution

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.10 | 0.25 | 0.35 | 0.20 | 0.10 |
| $F(x)$ | 0.10 | 0.35 | 0.70 | 0.90 | 1.00 |

The probabilities sum to 1, so the distribution is valid. $P(X \ge 3) = 0.20 + 0.10 = 0.30$; $P(1 \le X \le 3) = F(3) - F(0) = 0.90 - 0.10 = 0.80$.

## 2 Expectation, variance, standard deviation

**EXPECTED VALUE**

$$\mu = \mathrm{E}[X] = \sum_x x\, p(x). \tag{2.1}$$

This is the average value we would get over a very large number of repetitions (an average weighted by probabilities).

**VARIANCE AND STANDARD DEVIATION**

$$\sigma^2 = \mathrm{Var}(X) = \mathrm{E}\big[(X-\mu)^2\big] = \sum_x (x-\mu)^2 p(x) = \mathrm{E}\big[X^2\big] - \mu^2, \qquad \sigma = \sqrt{\mathrm{Var}(X)}. \tag{2.2}$$

The second form in (2.2) (the computational formula) is usually more convenient: $\mathrm{E}\big[X^2\big] = \sum x^2 p(x)$.

**EXAMPLE 2. CALLS: EXPECTATION AND SPREAD**

$\mu = 0 \cdot 0.10 + 1 \cdot 0.25 + 2 \cdot 0.35 + 3 \cdot 0.20 + 4 \cdot 0.10 = 1.95$. $\mathrm{E}\big[X^2\big] = 0.25 + 4 \cdot 0.35 + 9 \cdot 0.20 + 16 \cdot 0.10 = 0.25 + 1.40 + 1.80 + 1.60 = 5.05$. Hence $\mathrm{Var}(X) = 5.05 - 1.95^2 = 1.2475$, $\sigma = 1.117$.

**EXAMPLE 3. THE LOAN FROM LECTURE 3**

The profit $X$ (million UZS) takes the values $+1.2$, $+0.4$, $-8.0$ with probabilities $0.80$, $0.15$, $0.05$. Then $\mathrm{E}[X] = 0.62$, $\mathrm{E}[X^2] = 1.44 \cdot 0.8 + 0.16 \cdot 0.15 + 64 \cdot 0.05 = 1.152 + 0.024 + 3.2 = 4.376$, $\mathrm{Var}(X) = 4.376 - 0.62^2 = 3.9916$, $\sigma = 1.998$.
The expected profit is positive (0.62), but the standard deviation (2.0) is more than three times larger. The reason is the rare but heavy default: the "average" result is never observed on an individual loan (neither $0.62$ nor anything close to it is among the outcomes).

**COMMON MISTAKE**

The expected value need not be a possible value of the variable and does not guarantee the result of a single trial. It is a long-run average; on one loan you can lose 8.

## 3 Properties of expectation and variance

**Linear transformations and sums**

$$\mathrm{E}[aX+b] = a\mathrm{E}[X] + b, \qquad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X), \qquad \mathrm{SD}(aX+b) = |a|\,\mathrm{SD}(X),$$

$$\mathrm{E}[X+Y] = \mathrm{E}[X] + \mathrm{E}[Y] \text{ (always)},$$

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \text{ (if } X, Y \text{ are independent)}.$$

A shift $b$ changes the centre but not the spread; multiplication by $a$ scales the spread by $|a|$ and the variance by $a^2$. The expectation of a sum equals the sum of expectations even for dependent variables, whereas for the variance independence is essential (the general case is in Lecture 6).

**EXAMPLE 4. A PORTFOLIO OF 100 LOANS**

Let the profits of 100 loans be independent and identically distributed, as above. The total profit is $T = X_1 + \cdots + X_{100}$:

$$\mathrm{E}[T] = 100 \cdot 0.62 = 62, \qquad \mathrm{Var}(T) = 100 \cdot 3.9916 = 399.16, \qquad \mathrm{SD}(T) = 19.98.$$

The average profit per loan $\bar{X} = T/100$ has $\mathrm{E}[\bar{X}] = 0.62$ and $\mathrm{SD}(\bar{X}) = 19.98/100 = 0.20$, i.e. the spread of the mean is $\sqrt{100} = 10$ times smaller than that of a single loan. This is the seed of the law of large numbers and the CLT (Lecture 7). Note: standard deviations do not add, variances do.

**COMMON MISTAKE**

$\mathrm{SD}(X+Y) \neq \mathrm{SD}(X) + \mathrm{SD}(Y)$ and $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ (not the difference): subtracting independent variables does not reduce the spread.

# 4 The binomial distribution

The Bernoulli scheme: $n$ independent trials are performed, in each a "success" with probability $p$ (the same each time). The number of successes $X$ has the binomial distribution, $X \sim \text{Bin}(n, p)$.

**Binomial distribution**

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n; \qquad \text{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

Conditions of applicability (always check them): (1) a fixed number of trials $n$; (2) two outcomes; (3) the same probability of success; (4) independence. The factor $\binom{n}{k}$ is the number of ways to choose in which $k$ trials the success occurred (Lecture 2).

**EXAMPLE 5. QUALITY CONTROL**

The probability that a part is defective is 5%, a batch of 10 parts, $X \sim \text{Bin}(10, 0.05)$.

$$P(X = 0) = 0.95^{10} = 0.599,$$
$$P(X \geq 1) = 1 - 0.599 = 0.401,$$
$$P(X = 2) = \binom{10}{2} \cdot 0.05^2 \cdot 0.95^8 = 45 \cdot 0.0025 \cdot 0.6634 = 0.0746.$$

$\text{E}[X] = 0.5$, $\sigma = \sqrt{10 \cdot 0.05 \cdot 0.95} = 0.69$. In 40% of batches of 10 parts there is at least one defective, although "on average" there are only 0.5 defective parts.

**EXAMPLE 6. CUSTOMERS WHO ACCEPTED AN OFFER**

A bank calls 20 customers, each accepts with probability $0.3$ independently, $X \sim \text{Bin}(20, 0.3)$: $\text{E}[X] = 6$, $\text{Var}(X) = 4.2$, $\sigma = 2.05$. $P(X = 6) = \binom{20}{6} 0.3^6\, 0.7^{14} = 0.1916$; $P(X \leq 4) = 0.2375$; $P(X \geq 8) = 0.2277$. Such calculations use a table or the Excel function `BINOM.DIST`.

# 5 The geometric distribution

If trials are repeated until the first success, the number of trials $X$ has the geometric distribution:

$$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \dots; \qquad \text{E}[X] = \frac{1}{p}, \quad P(X \leq k) = 1 - (1-p)^k. \tag{5.1}$$

Example: the probability of closing a deal with one call is $p = 0.2$. Then the deal takes exactly 3 calls with probability $0.8^2 \cdot 0.2 = 0.128$, at most 3 calls with probability $1 - 0.8^3 = 0.488$, and on average $1/0.2 = 5$ calls are needed.

# 6 The Poisson distribution

If events occur independently of each other in continuous time (or space) with a constant average intensity, then the number of events $X$ in a fixed interval, $\lambda$ on average, has the Poisson distribution:

$$P(X=k)=\frac{e^{-\lambda}\lambda^{k}}{k!},\quad k=0,1,2,\dots;\qquad \mathrm{E}[X]=\mathrm{Var}(X)=\lambda. \tag{6.1}$$

**EXAMPLE 7. CUSTOMERS AT A BRANCH**

A branch receives on average $\lambda=3$ customers per 10 minutes. Then

$$P(X=0)=e^{-3}=0.0498,\quad P(X=2)=\frac{e^{-3}3^{2}}{2}=0.2240,$$
$$P(X\ge 4)=1-P(X\le 3)=0.3528.$$

The intensity is rescaled in proportion to the interval: in 20 minutes there are 6 customers on average, and, for example, $P(X\le 3)=0.1512$.

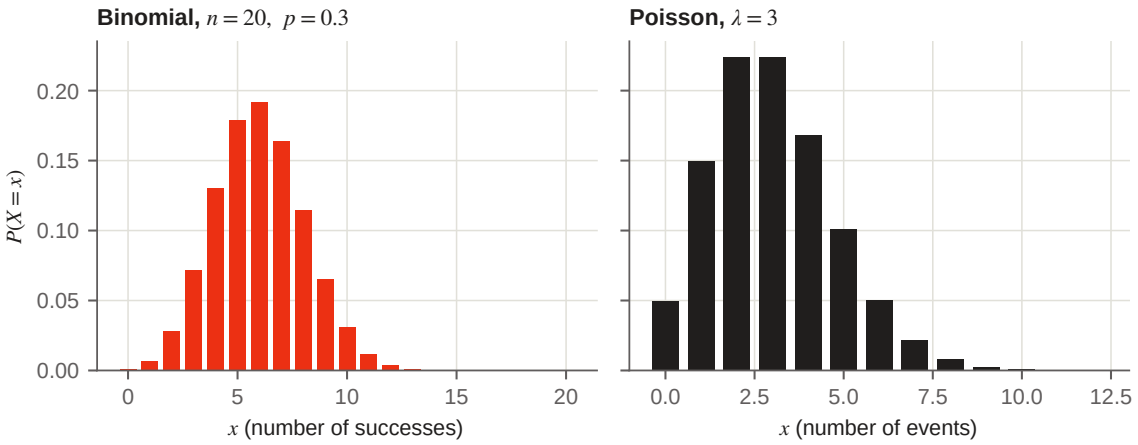



**Fig. 1.** Probability distributions: the binomial $\mathrm{Bin}(20,0.3)$ is almost symmetric and bell-shaped, centred at $\mu=6$; the Poisson with $\lambda=3$ is skewed to the right.

**Poisson as an approximation to the binomial.** For large $n$ and small $p$ the distribution $\mathrm{Bin}(n,p)$ is close to $\mathrm{Pois}(\lambda=np)$. For example, 200 operations, error probability $0.01$: $P(X\ge 3)=0.3233$ by the binomial formula and $0.3233$ by the Poisson with $\lambda=2$.

**REMEMBER**

How to choose a model: a fixed number $n$ of trials with success probability $p$ — binomial; the number of trials until the first success — geometric; the number of events in an interval with no upper bound — Poisson. For the Poisson the variance equals the expectation; if the spread of the data is noticeably larger than the mean, the model is inadequate.

# 7 Key terms

| Term | *In Russian* | Meaning |
|---|---|---|
| **Random variable** | *Случайная величина* | a numerical function of the outcome |
| **Probability mass function (PMF)** | *Закон распределения* | $p(x) = P(X = x)$, $\sum p = 1$ |
| **Cumulative distribution function (CDF)** | *Функция распределения* | $F(x) = P(X \le x)$ |
| **Expected value** | *Математическое ожидание* | $\mu = \sum x\, p(x)$ |
| **Variance** | *Дисперсия* | $\mathrm{E}[X^2] - \mu^2$ |
| **Binomial** | *Биномиальное* | $\mathrm{Bin}(n, p)$: $np$, $np(1-p)$ |
| **Geometric** | *Геометрическое* | number of trials until success, $1/p$ |
| **Poisson** | *Пуассона* | events in an interval, $\lambda$ |

**Formulas of Lecture 4**

$$\mu = \sum x\, p(x), \qquad \sigma^2 = \sum x^2 p(x) - \mu^2,$$
$$\mathrm{E}[aX + b] = a\mu + b, \quad \mathrm{Var}(aX + b) = a^2\sigma^2, \quad \mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \text{ (indep.)},$$
$$\mathrm{Bin}(n, p):\ P(k) = \tbinom{n}{k} p^k (1-p)^{n-k},\ \mu = np,\ \sigma^2 = np(1-p);$$
$$\mathrm{Geom}(p):\ P(k) = (1-p)^{k-1} p,\ \mu = 1/p; \qquad \mathrm{Pois}(\lambda):\ P(k) = e^{-\lambda}\lambda^k/k!,\ \mu = \sigma^2 = \lambda.$$

# 8 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

**Problem 1** [A] The distribution of the variable $X$: $P(1) = 0.2$, $P(2) = 0.3$, $P(3) = k$, $P(4) = 0.1$.

(a) Find $k$.
(b) Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$, $\mathrm{SD}(X)$.
(c) Find $P(X \ge 3)$.

**Problem 2** [A] For a variable $X$: $\mathrm{E}[X] = 40$, $\mathrm{SD}(X) = 6$. The variable $X_2$ is independent of $X$: $\mathrm{E}[X_2] = 25$, $\mathrm{SD}(X_2) = 8$.

(a) Find the expectation, variance and standard deviation of $Y = 3X - 10$.
(b) Find the expectation and standard deviation of $X + X_2$ and $X - X_2$.

**Problem 3** [A] $X \sim \text{Bin}(5, 0.4)$.

(a) Build the table of $P(X = k)$, $k = 0, \dots, 5$.
(b) Find $\text{E}[X]$ and $\text{Var}(X)$ by the formulas.
(c) Find $P(X \geq 4)$.

**Problem 4** [B] A borrower repays a loan with probability $0.9$ independently of the others. The bank issued 12 loans.

(a) Find the expected number of repaid loans and the standard deviation.
(b) Find the probability that all 12 are repaid.
(c) Find the probability that at most one loan is not repaid.

**Problem 5** [B] The average number of complaints to a support line is 1.5 per hour (a Poisson flow).

(a) Find the probability that there are no complaints in an hour.
(b) Find the probability of at least two complaints in an hour.
(c) Find the probability that at most one complaint arrives in two hours.

**Problem 6** [B] Two projects are considered. Project $A$: profit $+10$ million with probability $0.6$ and $-5$ million with probability $0.4$. Project $B$: $+6$ million with probability $0.7$ and $-1$ million with probability $0.3$.

(a) Find the expected profit and standard deviation of each project.
(b) Which project would you choose if the goal is to maximize expected profit? And if stability matters? Justify.

**Problem 7** [B] An insurance company sells a policy for 80 thousand UZS. Annual payouts on the policy: 0 with probability $0.94$, 400 thousand with probability $0.05$, 3000 thousand with probability $0.01$.

(a) Find the expected payout and the standard deviation of the payout on one policy.
(b) Find the expected profit and the standard deviation of profit on one policy.
(c) The company sold 1000 independent policies. Find the expected total profit and its standard deviation.

**Problem 8** [B] The profit of one loan has $\mu = 0.62$ million, $\sigma = 2.0$ million (Lecture 4, the loan example). The bank issues 25 independent loans. Find the expectation and standard deviation of the total profit and of the average profit per loan. How many times less variable is the average than one loan?

**Problem 9** [B] A bank processes 300 transactions per minute, each erroneous with probability $0.006$ independently.

(a) Find the probability of no errors by the exact binomial formula and by the Poisson approximation.
(b) Find the probability of at least three errors by both methods.

**Problem 10** [B] The probability of closing a deal on one call is $0.25$ (calls are independent). A manager calls until the first deal.

(a) What is the probability that the first deal comes on the fourth call?
(b) What is the probability that at most four calls suffice?

(c) How many calls are needed on average?

**Problem 11** [C] Prove: (a) $\operatorname{Var}(X) = \mathrm{E}[X^2] - \mu^2$; (b) $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$.

**Problem 12** [C] Let $X \sim \operatorname{Bin}(n, p)$ be the number of successes, $X = I_1 + \cdots + I_n$, where $I_i$ is the indicator of success in the $i$-th trial ($I_i = 1$ with probability $p$, otherwise 0, independent).

(a) Find $\mathrm{E}[I_i]$ and $\operatorname{Var}(I_i)$.
(b) Derive $\mathrm{E}[X] = np$ and $\operatorname{Var}(X) = np(1-p)$.
(c) At what $p$ is the variance of a single trial maximal? What is it equal to?